

JUN-GI JANG

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DGIST E7-214, 333, Techno Jungang Daero, Hyeonpung-Eup, Dalseong-Gun, Daegu, 42988, Korea

RESEARCH INTERESTS

Data Mining, Query-Aware Data Systems, Streaming and Adaptive Computation, Applied AI

EXPERIENCE

Assistant Professor Daegu Gyeongbuk Institute of Science & Technology (DGIST) SEP. 2025 - PRESENT

Postdoctoral Researcher University of Illinois Urbana-Champaign (UIUC) SEP. 2023 - AUG. 2025
Advisor: [Prof. Hanghang Tong](#)

Postdoctoral Researcher Seoul National University (SNU) MAR. 2023 - AUG. 2023
Advisor: [Prof. U Kang](#)

Research Intern HYPERCONNECT JUL. 2020 - AUG. 2020

EDUCATION

Seoul National University MAR. 2017 - FEB. 2023
Ph.D. in Computer Science and Engineering
Thesis: Mining Real World Tensors via Efficient Tensor Decomposition Methods
Advisor: [Prof. U Kang](#)

Seoul National University MAR. 2010 - FEB. 2017
B.S. in Mechanical and Aerospace Engineering;
and Computer Science and Engineering (double major)

BEST PAPER AWARDS

Best Paper Award (Honorable Mention), ICDE MAY 2022

Best Paper Award (Best Research Paper), KDD AUG. 2021

Best Paper Award (1st Place), Bigcomp JAN. 2021

OTHER AWARDS AND HONORS

Outstanding Dissertation Award, SNU CSE FEB. 2023

SNU BK21 Star Researcher Award, SNU BK21 FEB. 2022

BK21 Best Graduate Student Award, SNU BK21 FEB. 2022

Future Gauss Lecture Award, Gauss Labs FEB. 2022

Naver Ph.D. Fellowship Award, Naver DEC. 2021

Qualcomm Innovation Fellowship, Qualcomm NOV. 2021

Yulchon AI Star Fellowship, Yulchon Foundation SEP. 2021

Humantech Paper Award (Honorable Mention, lead-author), Samsung FEB. 2018

REFEREED CONFERENCE AND JOURNAL PAPERS

28. **SCOUT: Coupling-Free Bounds for Trillion-Scale Top-k Retrieval in Sparse Tensor Factorization**
Jun-Gi Jang, Hyunsik Yoo, Ruizhong Qiu, Yinglong Xia, and Hanghang Tong.
ACM SIGMOD/PODS International Conference on Management of Data (**SIGMOD**), 2027, Huntington Beach, CA, USA. To Appear

27. **ProgNet: Program-Grounded Evidence Composition for Interpretable Graph Classification**
Minseok Jeon, Seunghyun Park, and **Jun-Gi Jang**.
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea. To Appear
26. **DuET: Dual-View Tensor-to-Topology Spectral Adapter for Enhancing Sparse Tensor Factorization**
Jun-Gi Jang, Jingrui He, Andrew J Margenot, Hanghang Tong.
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea. To Appear
25. **Fast and Accurate Domain Adaptation for Irregular and Regular Tensor Decomposition**
Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang.
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2026
24. **TUCKET: A Tensor Time Series Data Structure for Efficient and Accurate Factor Analysis over Time Ranges**
Ruizhong Qiu*, **Jun-Gi Jang***, Xiao Lin, Lihui Liu, Hanghang Tong (* equal contribution)
International Conference on Very Large Data Bases (**VLDB**), 2025, London, United Kingdom.
23. **Improving Group Fairness in Tensor Completion via Imbalance Mitigating Entity Augmentation**
Dawon Ahn*, **Jun-Gi Jang***, and Evangelos E. Papalexakis (* equal contribution)
The Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**), 2025, Sydney, Australia
22. **Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors**
Jun-Gi Jang, Yong-chan Park, and U Kang
ACM International Conference on Information and Knowledge Management (**CIKM**), 2024, Boise, Idaho, USA.
Oral presentation, acceptance rate $347/1496 \approx 23\%$.
21. **Compact Lossy Compression of Tensors via Neural Tensor-Train Decomposition**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, Kijung Shin
Knowledge and Information Systems (**KAIS**), Oct., 2024
20. **Compact Decomposition of Irregular Tensors for Data Compression: From Sparse to Dense to High-Order Tensors**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, and Kijung Shin
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
Oral presentation, acceptance rate $\approx 20\%$.
19. **Fast and Accurate Domain Adaptation for Irregular Tensor Decomposition**
Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
Oral presentation, acceptance rate $\approx 20\%$.
18. **Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors - Algorithm and Application**
Jun-Gi Jang, Jeongyoung Lee, Yong-chan Park, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2023, Long Beach, CA, USA
Oral presentation, acceptance rate $313/1416 \approx 22.1\%$.
17. **Accurate Open-set Recognition for Memory Workload**
Jun-Gi Jang, Sooyeon Shim, Vladimir Egay, Jeeyong Lee, Jongmin Park, Suhyun Chae, and U Kang

ACM Transactions on Knowledge Discovery from Data (TKDD), June, 2023

16. **Fast and accurate interpretation of workload classification model**
Sooyeon Shim, Doyeon Kim, **Jun-Gi Jang**, Suhyun Chae, Jeeyong Lee, and U Kang
PLOS ONE, March, 2023
15. **Accurate Bundle Matching and Generation via Multitask Learning with Partially Shared Parameters**
Hyunsik Jeon, **Jun-Gi Jang**, Taehun Kim, and U Kang
PLOS ONE, March, 2023
14. **Static and Streaming Tucker Decomposition for Dense Tensors**
Jun-Gi Jang and U Kang
ACM Transactions on Knowledge Discovery from Data (TKDD), Feb., 2023
13. **Falcon: Lightweight and Accurate Convolution Based on Depthwise Separable Convolution**
Jun-Gi Jang*, Chun Quan*, Hyun Dong Lee, and U Kang
Knowledge and Information Systems (KAIS), Jan., 2023 (* equal contribution)
12. **Accurate PARAFAC2 Decomposition for Temporal Irregular Tensors with Missing Values**
Jun-Gi Jang, Jeongyoung Lee, Jiwon Park, and U Kang
IEEE International Conference on Big Data (BigData), 2022, Osaka, Japan
Oral presentation, acceptance rate $122/633 \approx 19.2\%$.
11. **DPar2: Fast and Scalable PARAFAC2 Decomposition for Irregular Dense Tensors**
Jun-Gi Jang and U Kang
IEEE International Conference on Data Engineering (ICDE) 2022, Virtual Event
Oral presentation, acceptance rate $211/780 \approx 27.1\%$
🏆 **Best Paper Award, Honorable Mention**
10. **Large-scale tucker Tensor factorization for sparse and accurate decomposition**
Jun-Gi Jang*, Moonjeong Park*, Jongwuk Lee, and Lee Sael
The Journal of Supercomputing, May, 2022. (* equal contribution).
9. **Finding Key Structures in MMORPG Graph with Hierarchical Graph Summarization**
Jun-Gi Jang, Chaeheum Park, Changwon Jang, Geonsoo Kim, and U Kang
ACM Transactions on Knowledge Discovery from Data (TKDD), Feb., 2022
8. **Time-Aware Tensor Decomposition for Sparse Tensors**
Dawon Ahn, **Jun-Gi Jang**, and U Kang
Machine Learning, Sep., 2021
7. **Fast and Memory-Efficient Tucker Decomposition for Answering Diverse Time Range Queries**
Jun-Gi Jang and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2021, Virtual Event
Oral presentation, acceptance rate $238/1541 \approx 15.4\%$
🏆 **Best Paper Award, Best Research Paper**
6. **Fast and Accurate Partial Fourier Transform for Time Series Data**
Yong-chan Park, **Jun-Gi Jang**, and U Kang
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2021, Virtual Event
Oral presentation, acceptance rate $238/1541 \approx 15.4\%$
5. **VEST: Very Sparse Tucker Factorization of Large-Scale Tensors**
Moonjeong Park*, **Jun-Gi Jang***, and Lee Sael

IEEE International Conference on Big Data and Smart Computing (**BigComp**), 2021, Online (* equal contribution)

🏆 **Best Paper Award, 1st Place**

4. **D-Tucker: Fast and Memory-Efficient Tucker Decomposition for Dense Tensors**

Jun-Gi Jang and U Kang

IEEE International Conference on Data Engineering (**ICDE**), 2020, Online

Short, acceptance rate $\approx 32\%$

3. **S3CMTF: Fast, accurate, and scalable method for incomplete coupled matrix-tensor factorization**

Dongjin Choi, **Jun-Gi Jang**, and U Kang

PLOS ONE, June, 2019.

2. **High-Performance Tucker Factorization on Heterogeneous Platforms**

Sejoon Oh, Namyong Park, **Jun-Gi Jang**, Lee Sael, and U Kang

IEEE Transactions on Parallel and Distributed Systems (**TPDS**), Apr., 2019

1. **Zoom-SVD: Fast and Memory Efficient Method for Extracting Key Patterns in an Arbitrary Time Range**

Jun-Gi Jang, Dongjin Choi, Jinhong Jung, and U Kang

ACM International Conference on Information and Knowledge Management (**CIKM**), 2018, Lingotto, Turin, Italy

Oral presentation, acceptance rate $147/826 \approx 17.8\%$

RESEARCH GRANTS

Young Scientist Grants (Type B), NRF of South Korea

MAR. 2026 - FEB. 2031

Postdoctoral Fellowship Program, NRF of South Korea

SEP. 2023 - AUG. 2024

TEACHING EXPERIENCE

DGIST

Instructor, BE202 Introduction to Data Science @ DGIST

SPRING 2026 **Co-Instructor**, TM545c

Fundamentals of Data Science @ DGIST

SPRING 2026

Seoul National University

Lead T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU

SPRING 2020

T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU

FALL 2019

T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data @ SNU

SPRING 2019

T.A., M1522.001400 Introduction to Data Mining @ SNU

SPRING 2018

T.A., M1522.000900 Data Structure @ SNU

FALL 2017

INVITED TALKS

Data Intelligence Workshop, DASFAA,

APR. 2026

The Future of Data Workshop 2023, KCC DB Society, KIISE

JUN. 2023

Korea Computer Congress 2022, KIISE

JUN. 2022

Korea Software Congress 2021, KIISE

DEC. 2021

Regular Seminar, Qatar Computing Research Institute (QCRI)

SEP. 2021

Korea Computer Congress 2020, KIISE

JUL. 2020

Korea Software Congress 2018, KIISE

DEC. 2018

Samsung AI Forum, Samsung

SEP. 2018

PROFESSIONAL SERVICES

Workshop Organizing Committees

Workshop on Interplay Between Classical Tensor Methods And Foundation Models	2026
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Area Chair

KDD	2026
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Session Chair

DASFAA	2026
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PC Member and Reviewer

KDD	2023 - 2026
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CIKM	2025 - 2026
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WWW	2026
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AAAI	2024 - 2026
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SDM	2024
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KAIS journal	2024
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TKDE journal	2024
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TSP journal	2024
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Machine Learning journal	2023 - 2024
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TIST journal	2023
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TPDS journal	2023
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DAMI journal	2023
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External Reviewer

TKDD	2025
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NAACL	2024
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DSAA	2024
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KDD	2019 - 2022
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WWW	2019 - 2021
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ICLR	2021
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NeurIPS	2020 - 2022
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CIKM	2018 - 2019
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ICDM	2018
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WSDM	2018
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